KOZE - Sheep

#bfs (/problems/tag/bfs)

Mirko has a herd of sheep, surrounded by fences backyard. While he was asleep, wolves have sneaked into the fenced area and attacked the sheep.

Backyard is of a rectangular shape, and consists of fields arranged in rows and columns.

- Character '**.**' (full stop) represents a blank field.
- Character '**#**' represents a fence.
- Character '**k**' represents a sheep.
- Character '**v**' represents a wolf.

Two fields belong to the same sector if we can move from the field **A** to the field **B** without going over the fence, by making only **horizontal** and **vertical** steps (we cannot move diagonally).

If we can escape from field A from the backyard, that field does not belong to any sector.

Luckily, Mirko taught his sheep Kung-Fu skills, and **they can defend themselves against wolves only if they outnumber the wolves in that sector**. When there are more sheep in the sector than wolves, all wolves die without sheep casualties. Otherwise all sheep perish and wolves are unharmed. If a field doesn't belong in any sector, sheep will flee and wolfs will be left without a prey, so every animal survives.

Write a program that will determine how many sheep and wolves will survive this bloody night.

Input

Integers N and M, number of rows and columns which represent Mirko's backyard.

In every of the N lines, there are M characters representing the appearance of Mirko's backyard - positions of the fences, wolves and sheep.

Constraints

$3 \leq N, M \leq 250$

Output

In the first and the only line, print the number of **sheep** and **wolves** that will survive.